

DASC7606 - 6B

Deep Learning

Natural Language Processing

RNN/LSTM – word embedding

Dr. Bethany Chan

Professor Francis Chin

2025

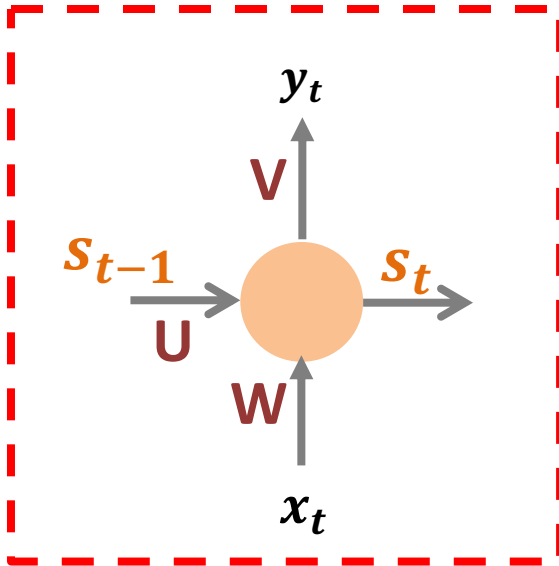
NLP – Outline

- Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTMs)
- Word embeddings (representations)
 - CBOW, Skip-gram
 - Evaluation
- Machine Translation, Encoder-Decoder models, Seq2seq

Recurrent Neural Networks (RNN)

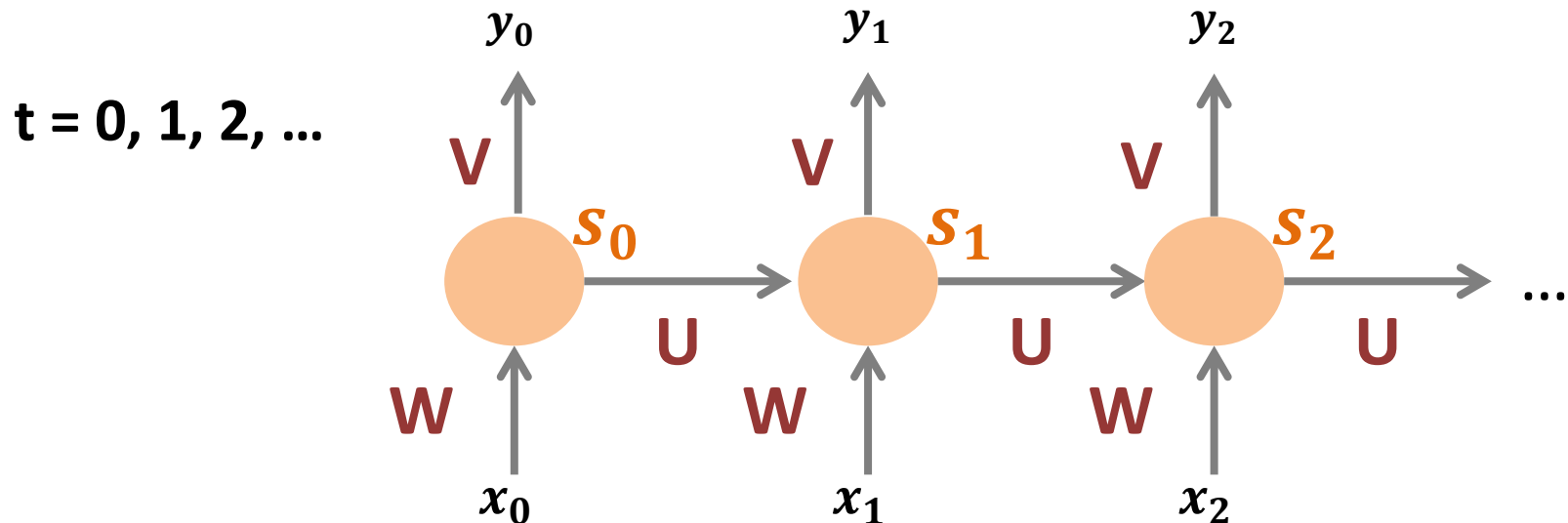
- Simpler versions of NN with fewer weights to learn (like CNN)
- Input of RNN is usually in the form of a sequence $x_0, x_1, \dots, x_n$ where **order is important**
- For sentences in natural language processing, order is important, for example:
 - The food is good, not bad at all.
 - The food is bad, not good at all.

Recurrent Neural Network (RNN)



x_t : input at time t
 s_t : state at time t
 y_t : output, if any, at time t
 U, V, W : weight matrices

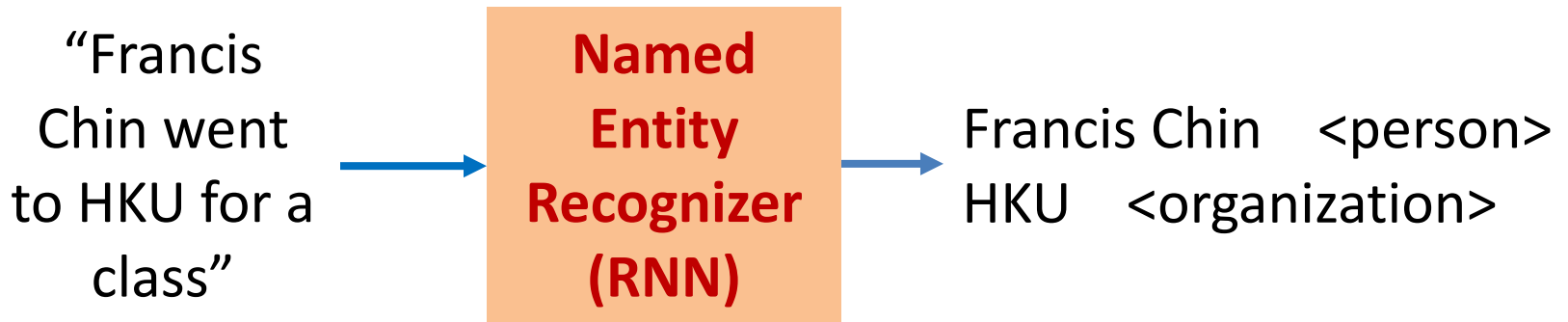
$$s_t = g(Wx_t + Us_{t-1})$$
$$y_t = g(Vs_t)$$



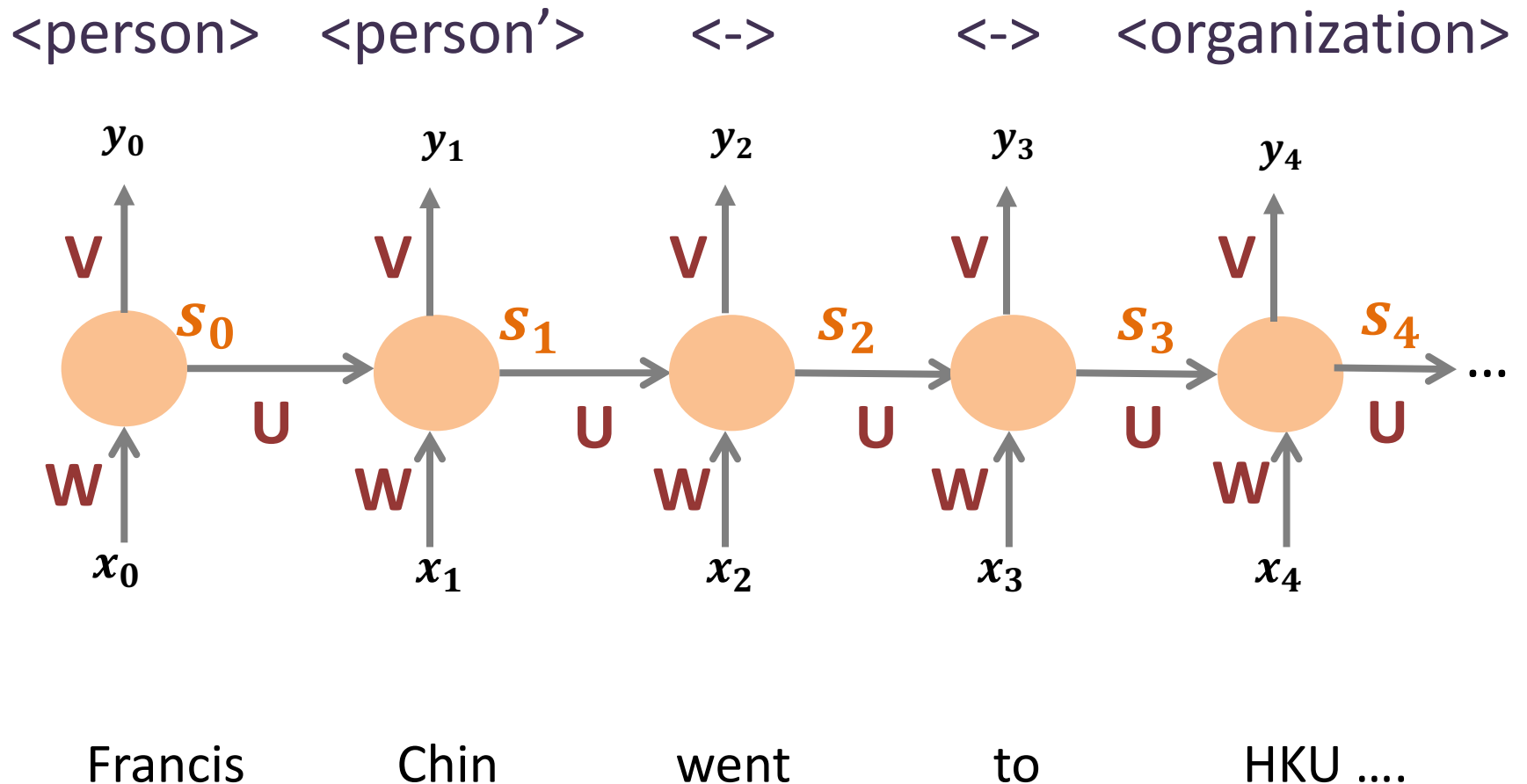
RNN: Pros and Cons

Advantages	Drawbacks
Possibility of processing input of any length	Computation being slow
Weights (U, V, W) are shared across time, model size does not increase with size of input	Cannot consider any future input for the current state [will try to mitigate]
Computation takes into account historical information	Difficulty of accessing information from a long time ago [will try to mitigate]

Application: Named Entity Recognizer



Application: Named Entity recognizer



Application: Sentiment Analyzers

“the food is
good and
the service
is excellent”



**Sentiment
Analyzer
(RNN)**



Positive

“the place is
dirty and
noisy”



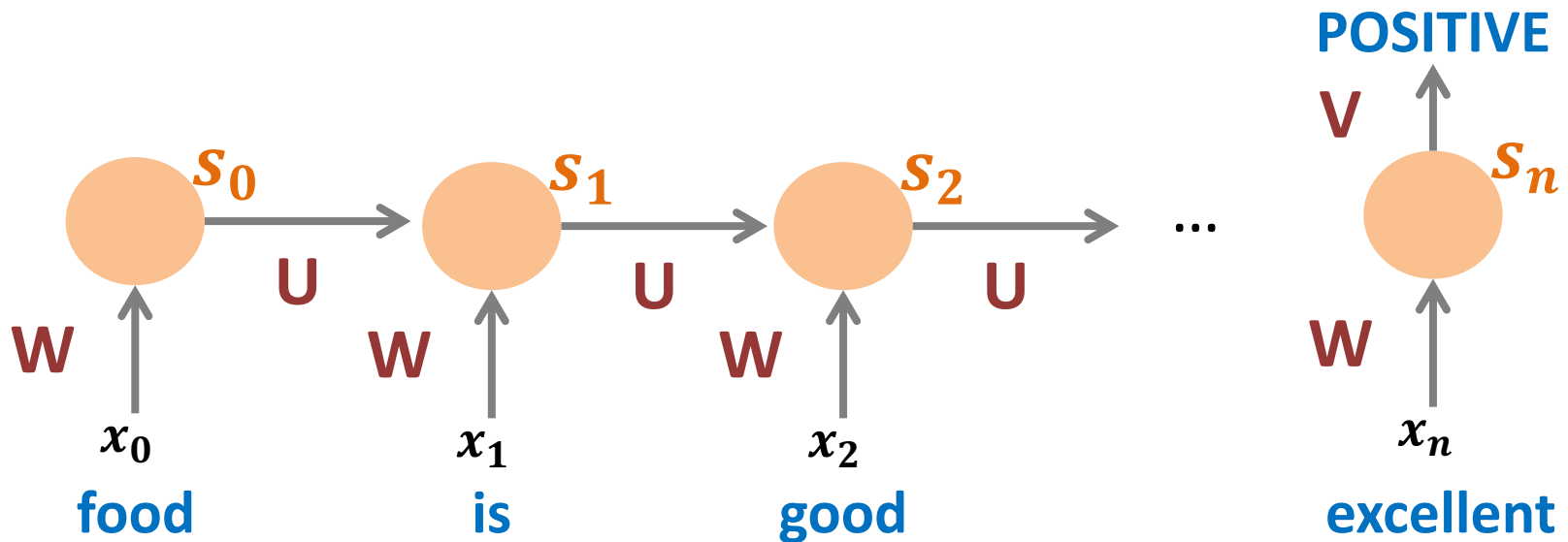
**Sentiment
Analyzer
(RNN)**



Negative

Application: Sentiment Analysis

- Classification
- To understand the tone of a text
happy, sad, angry, disappointed, surprised, proud, scared,
“like”, “don’t like”,...



Application: Language Models

All work of
Shakespeare



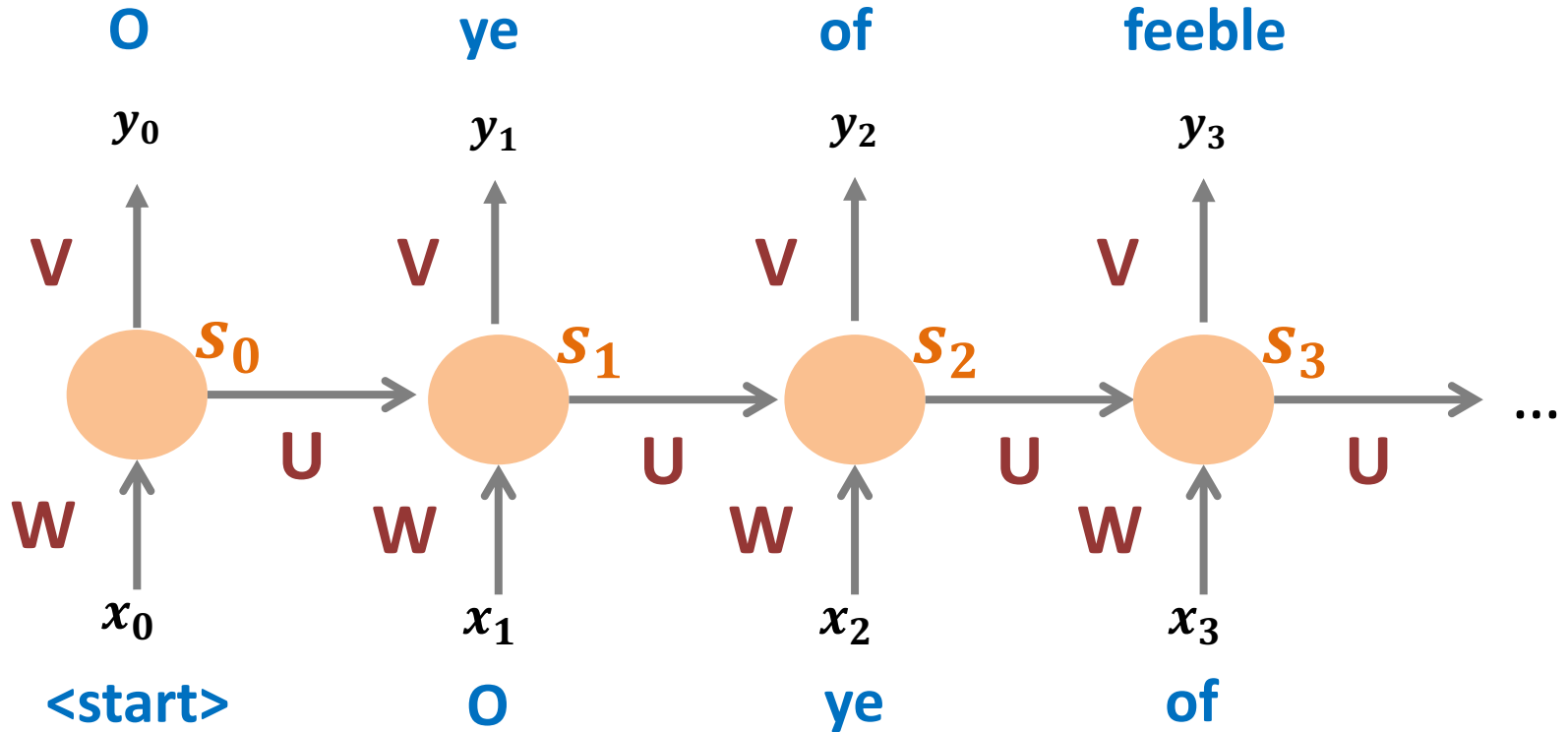
**Language
Model
(RNN)**



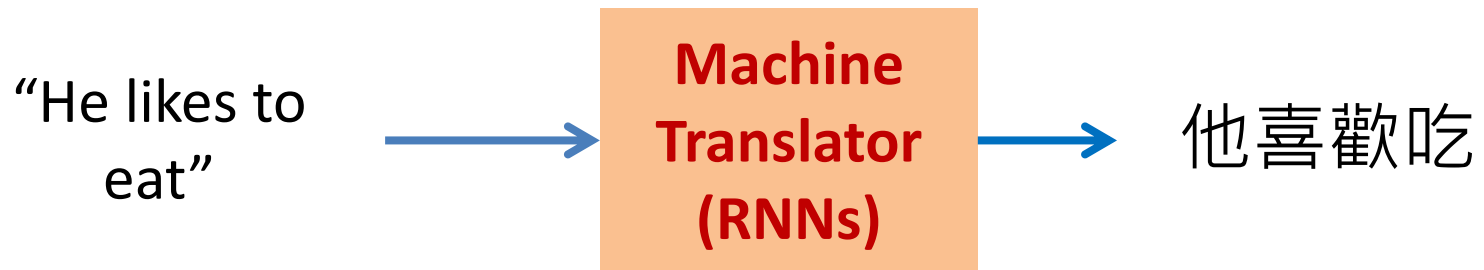
KING LEAR:

O, if you were a feeble sight, the
courtesy of your law,
Your sight and several breath, will
wear the gods
With his heads, and my hands are
wonder'd at the deeds,
So drop upon your lordship's head,
and your opinion
Shall be against your honour.

Application: Language Models



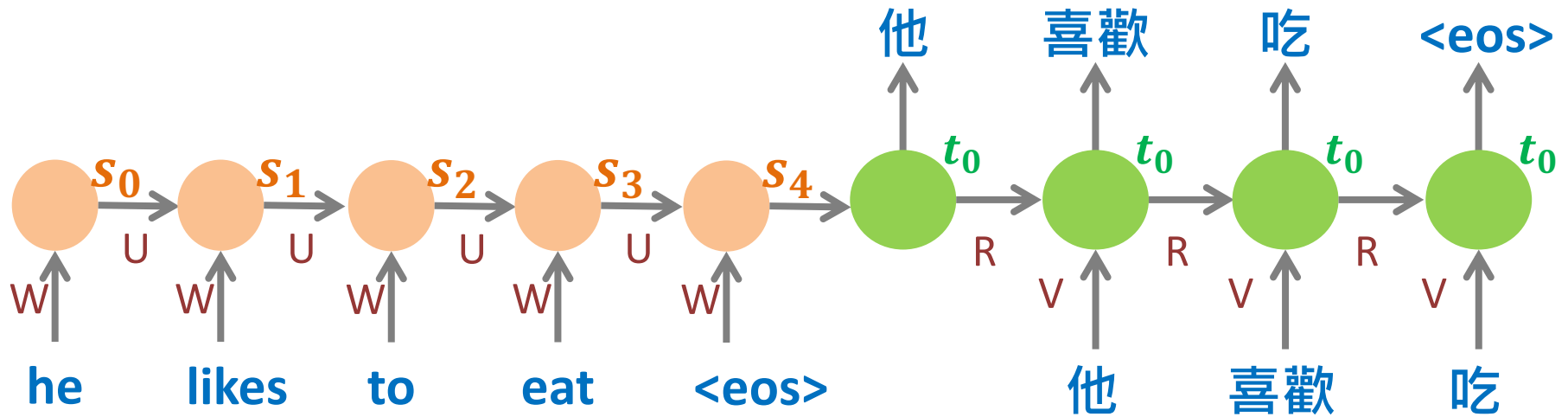
Application: Machine Translation



Application: Machine Translation

ENCODER

DECODER



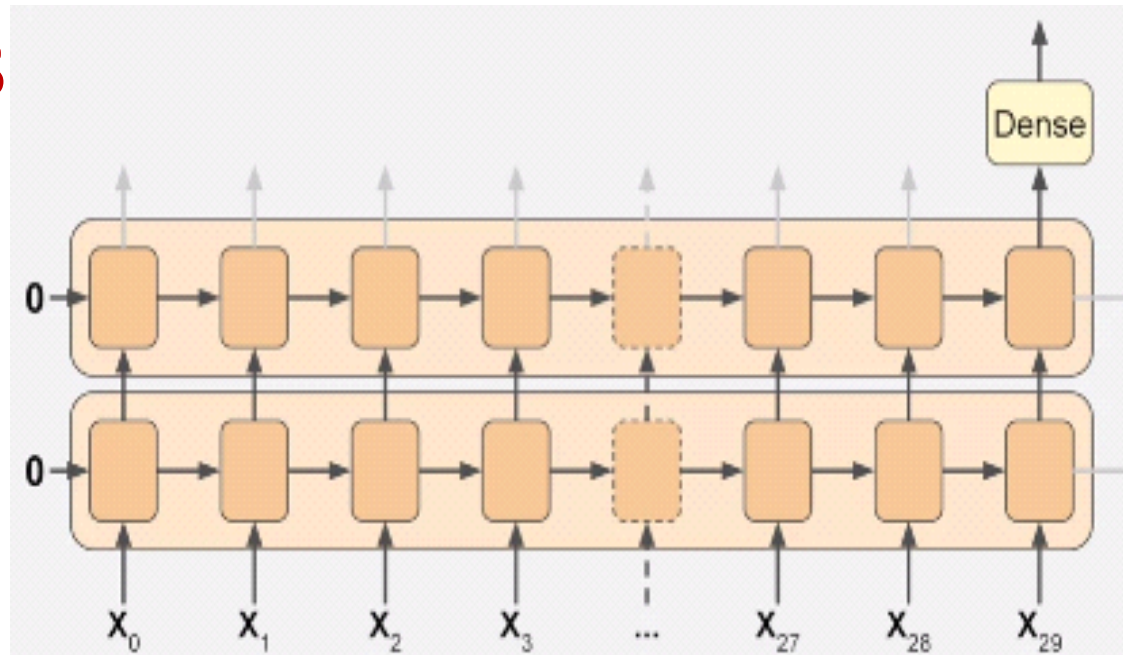
This application will be discussed in detail later as the Encoder-Decoder framework can lead to many other applications .

RNN: Summary of Applications

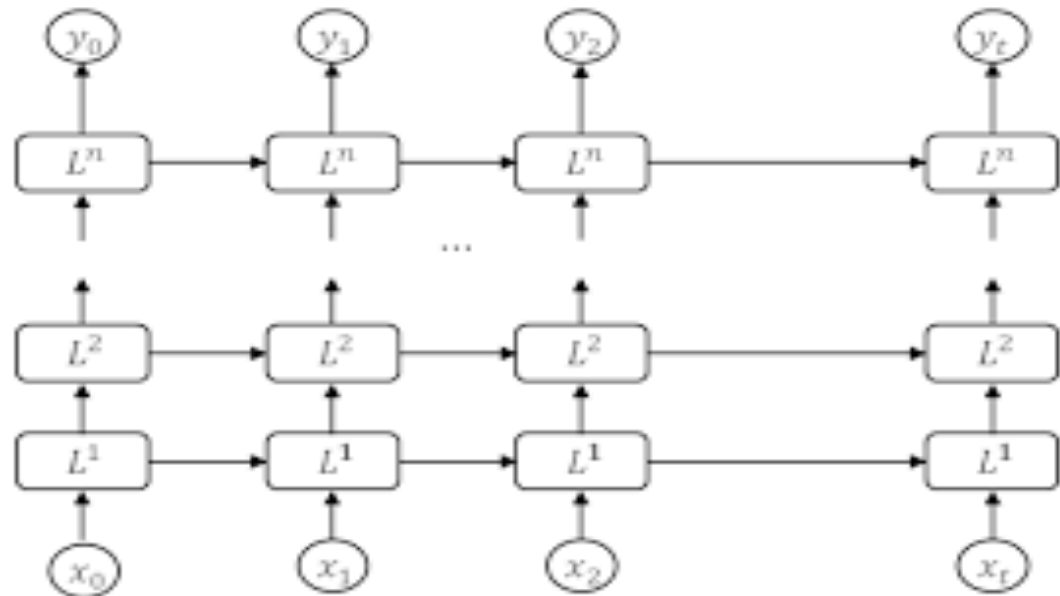
Types of RNN	Illustration	Example
One-to-Many $T_x = 1, T_y > 1$		Language model
Many-to-One $T_x > 1, T_y = 1$		Sentiment Classification
Many-to-Many $T_x = T_y$		Named Entity Recognition (NER)
Many-to-Many $T_x \neq T_y$		Machine Translation

RNN: Variants

2-layer RNN



Multi-layer RNN



Forget Problem and LSTM

Fill in the blank questions ...

Example 1:

Grace was hungry. The cookies were eaten by ____.

Example 2:

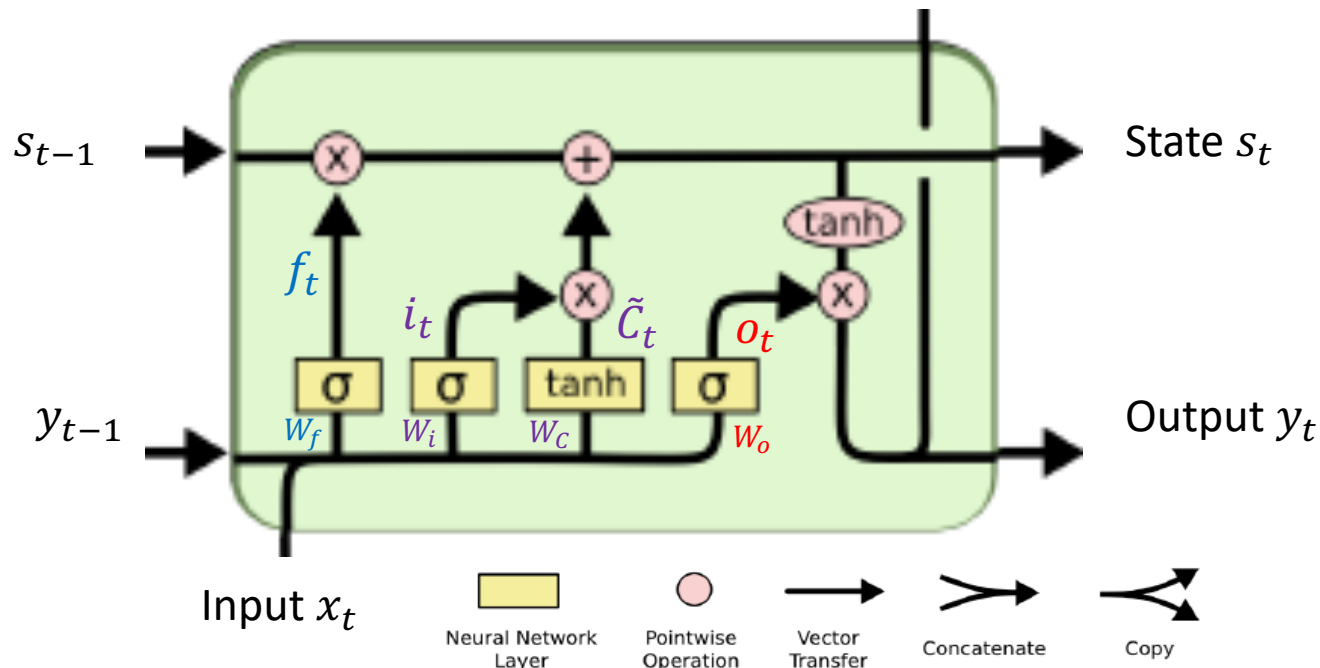
Grace was hungry. Beth brought some cookies home, and she was not hungry. Francis could not find the cookies. The cookies were eaten by ____.

Catering to Long-Term Dependencies

- Simple RNN cell → more complex gated cell
- **Long Short-Term Memory (LSTM)**
 - **Forget** irrelevant parts of the previous state (e.g. forget gender)
 - Selectively **update** cell state values (e.g. update new gender)
 - **Output** certain parts of cell state (e.g. “she”, “he”)

Ref: <http://colah.github.io/posts/2015-08-Understanding-LSTMs/>

Long Short-Term Memory (LSTM)



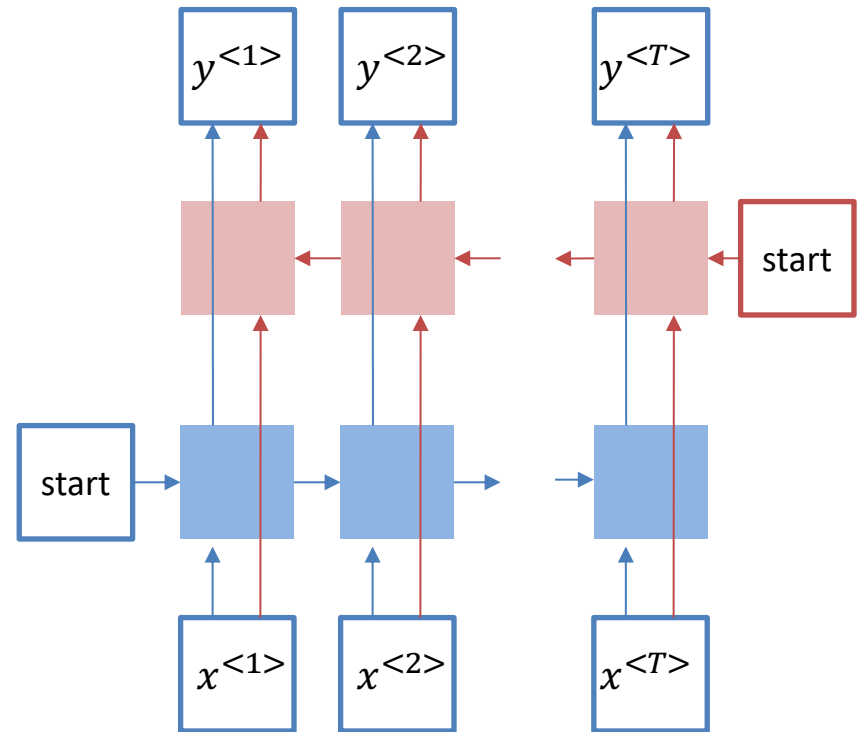
Purpose of LSTM equations	Equations for LSTM	Equations for RNN
Forget irrelevant parts of previous state	$f_t = \sigma(W_f \cdot [y_{t-1}, x_t])$	
Selectively update state	$i_t = \sigma(W_i \cdot [y_{t-1}, x_t])$ $\tilde{C}_t = \tanh(W_C \cdot [y_{t-1}, x_t])$ $s_t = f_t * s_{t-1} + i_t * \tilde{C}_t$	$s_t = g(Wx_t + Us_{t-1})$
Output certain parts of state	$o_t = \sigma(W_o \cdot [y_{t-1}, x_t])$ $y_t = o_t * \tanh(s_t)$	$y_t = g(Vs_t)$

RNNs: Bidirectional RNN

Consider the following

- I am _____.
- I am _____ hungry.
- I am _____ hungry,
and I can eat half a pig.

The end of the phrase
(if available) conveys
significant information
about which word to pick.



Information from both ends of the sequence is used to estimate the output. That is, we use information from both future and past observations to predict the current one.

Word Embedding

How are words represented?

Local representation (One-hot representation):

One token for each word as an N-dim vector

man = (0, 0, ..., 0, **1**, 0, 0, ..., 0, 0)

woman = (0, 0, ..., 0, ...0, **1**, 0, 0, ...0)

human = (0, 0, ..., **1**, ...0, 0, 0, 0, ...0)

manly = (0, 0, ..., 0, 0, 0, 0, **1**, 0, ... 0)

- shortcomings:
 - Size of space, R^N
(size of vocabulary, N = hundred thousand English words)
 - semantic independent (each word is independent)

Word Embedding: Distributed

Each word as an N-dim vector

king	=	(0, 1, ..., 1, ...1, 0,75, ..0)
queen	=	(0, 0, ..., 1, ...0, 1,67, ..0)
prince	=	(1, 0, ..., 1, ...1, 1,46, ..1)
princess	=	(0, 0, ..., 1, ...0, 0,42, ..0)

royalty sex age

Advantages:

- More elements can be represented 2^N (even more for real numbers)
- A bit at a position participates in representing multiple numbers (capturing some commonality between different words– **distributed**)
e.g., last bit =1 of a binary number means odd numbers

Word Embedding transform words into **Word Vectors** which encode similarity, difference, semantic relationship between words and the context of different words as used by human.

- **Word vector “embedding”** is a point in a multi-dimensional space, where each dimension represents a particular **feature** of the word
- **DL learns both features and corresponding values (real numbers) such that similar words mapped to nearby vectors.**
- How do we come up with word embeddings? features and values?
(by deep learning, end2end learning)

Word Embedding: Similarity & Context

Distributional hypothesis (linguistics) Harris (1954) Firth (1957):

- words that occur in similar contexts tend to have similar meanings
- you shall know a word by the company it keeps

Word-Document matrix

Terms	Doc 1	Doc 2	Doc 3
cat	2	3	0
dog	2	2	0
bank	0	0	2
pet	4	4	0
cash	0	0	2

No of times the word
“pet” appears in
document 1

Window-based co-occurrence matrix

	cat	dog	bank	pet	cash	eat
cat	-	3	0	4	0	2
dog	3	-	0	5	0	2
bank	0	0	-	1	5	0
pet	4	5	1	-	2	1
cash	0	0	5	2	-	0
eat	2	2	0	1	0	-

No of times the word “cash” and
word “bank” appear together within
a window of fixed size in a document.